

AUGUST 2023

Question 2

- a. A company you work for is required to design a memory block of size 512x16, however, due to financial constraints your company can only develop memory chips of size 128x8. As an engineer show how you can design the required memory block using the available units.

Solution

Step 1: Determine the number of chips required.

- a) Increase the word size

Required word size = 16bits

Available chip word size = 8bits

Therefore,

$$16 / 8 = 2$$

So, 2 chips are connected in parallel to form on **128 x 16** memory module.

- b) Increase the memory depth

Required locations = 512

Available locations = 128

Therefore,

$$512 / 128 = 4$$

So, **4 banks** of **128 x 16** memory are required.

- c) Total number of chips

Each bank requires 2 chips, and there are 4 banks.

$$4 \times 2 = 8 \text{ chips}$$

Answer: 8 memory chips (128 x 8 each) are required.

Step 2: Address lines

For a **512 x 16** memory:

$$512 = 2^9$$

Hence, **9 address lines** are required:

A0 to A8

Each 128 x 8 chip has:

$$128 = 2^7$$

So it requires **7 address lines**:

A0 to A6

The remaining 2 address lines (**A7 and A8**) are used to select one of the 4 memory banks using a **2-to-4 decoder**.

Step 3: Connections

Address Lines

- A0 – A6 connect to all chips
- A7 and A8 connect to a 2-to-4 decoder

Decoder outputs:

A8	A7	Selected Bank
0	0	Bank 0
0	1	Bank 1
1	0	Bank 2
1	1	Bank 3

Each decoder output enables 2 chips in one bank

Data lines

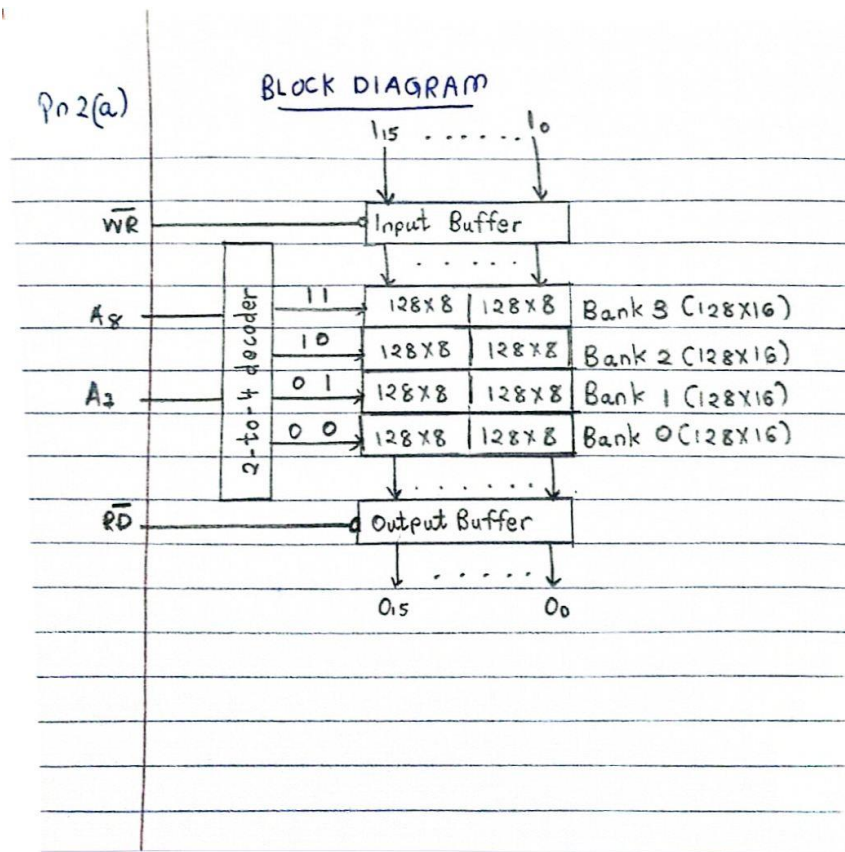
Within each bank:

- Chip 1 -> Data bits D0 – D7
- Chip 2 -> Data bits D8 – D15

Control signals

The Read (RD) and Write (WR) signals are connected to all chips, while only the enabled bank responds because its chip select (CS) is activated by the decoder.

Step 4: Block Diagram



Final Answer

To construct a 512×16 memory block using 128×8 memory chips:

- Number of chips required: 8
- Organization: 4 banks of 128×16
- Each bank: Two 128×8 chips connected in parallel
- Address lines: 9 ($A_0 - A_8$)
 - o $A_0 - A_6$ connect to every chip
 - o $A_7 - A_8$ feed a 2-to-4 decoder to select one of the 4 banks
- Data bus: 16 bits ($D_0 - D_{15}$)
- Control signals: $\overline{RD}$ and $\overline{WR}$ connect to all chips; the decoder-generated chip-select signal enables only the active bank.

b. Describe the 3 functional bus of the Intel 8085 microprocessor.

i. Address bus

It is a group of 16-bit lines identified as A_0 to A_{15} .

It is used to identify specific memory locations or peripheral I/O ports during read and write operations.

A15 – A8 are unidirectional and carry the high-order address while A7 – A0 carry the low-order address.

ii. Data bus

It is a group of 8 lines (AD7 to AD0) used for data flow. It is bidirectional.

It enables the MPU to manipulate an 8-bit data ranging from 00H – FFH.

iii. Control bus

It is comprised of various single lines that carry synchronization signals that control operations and identify devices with which the MPU is communicating.

c. Describe the function of the following Intel 8085 signals:

- i. **INTA** (Interrupt Acknowledge) – It is used to acknowledge an INTR (Interrupt Request) interrupt.
- ii. **TRAP** – It is a non-maskable interrupt that has the highest priority, reserved for critical system emergencies, such as sudden power failures or catastrophic system hardware errors.
- iii. **HOLD** – It is the signal that indicates that a peripheral such as DMA (Direct Memory Access) is requesting the use of the address and data buses.
- iv. **READY** – It is used by the microprocessor to sense whether a peripheral is ready or not for data transfer. If not, the processor waits. It is thus used to synchronize slower peripherals to the microprocessor.

- d. To reduce the complexity in designing the microprocessor, some control signals are generated by taking a logical combination of primary signals. On that note explain how the following signals are generated.

i. MEMR

When $RD = 0$, $WR = 0$ and $IO/M = 0$, memory read becomes active. Selected memory chip places data onto $D0 - D7$.

$MEMR \text{ (active-low)} = IO/M + RD \text{ (OR gate)}$

ii. IOR

Generated when $RD = 0$, $WR = 0$, $IO/M = 1$.

$IOR \text{ (active low)} = (IO/M)' + RD$

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Question 2

- a. Given the address bus structure of the Intel 8085. Describe how the bus structure determines the memory size of the microprocessor.

The Intel 8085 microprocessor has a 16-bit address bus consisting of address lines $A0 - A15$. Each address line can be either 0 or 1, allowing the processor to generate $2^{16} = 65,536$ unique memory addresses.

Since each address corresponds to 1 byte (8 bits) of memory, the maximum addressable memory is 65,536 bytes (64KB). Therefore, the maximum memory that the Intel 8085 can address is 65,536 bytes = 64KB of memory.

- b. With the aid of a diagram, describe the function of each flag in the flag register of the Intel 8085 microprocessor.

Zero (Z) flag -> set to 0 when an arithmetic operation results in 0

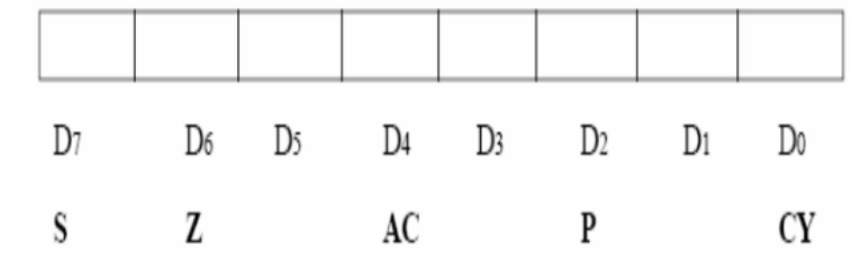
Carry (CY) flag -> set to 1 if sum in the accumulator is larger than 8bits.

Sign (S) flag -> set to 1 if the result is negative or the left most bit is 1

Parity (P) flag -> set to 1 if there is an even number of 1s.

Auxiliary Carry (AC) flag -> set to 1 if there is a carry from the lower nibble.

Diagram



- c. List any TWO 16-bit registers of the Intel 8085 microprocessor.

Stack Pointer

Program Counter

- d. Two 8Kx8 EPROMs are to be interfaced with the Intel 8085 where address lines A15 – A13 are used for chip select through 3-to-8 decoder. Design the memory interfacing required in this case.

Step 1: Analyze the memory size.

Each EPROM is 8K x 8.

$8K = 8192 = 2^{13}$.

Therefore, 13 address lines are required for each EPROM:

A0 – A12

The Intel 8085 has 16 address lines (A0 – A15).

So,

- A0 – A12 -> connected to both EPROMS
- A13 – A15 -> connected to the 3-to-8 decoder for chip selection. Outputs are Y0 and Y1

Memory Map

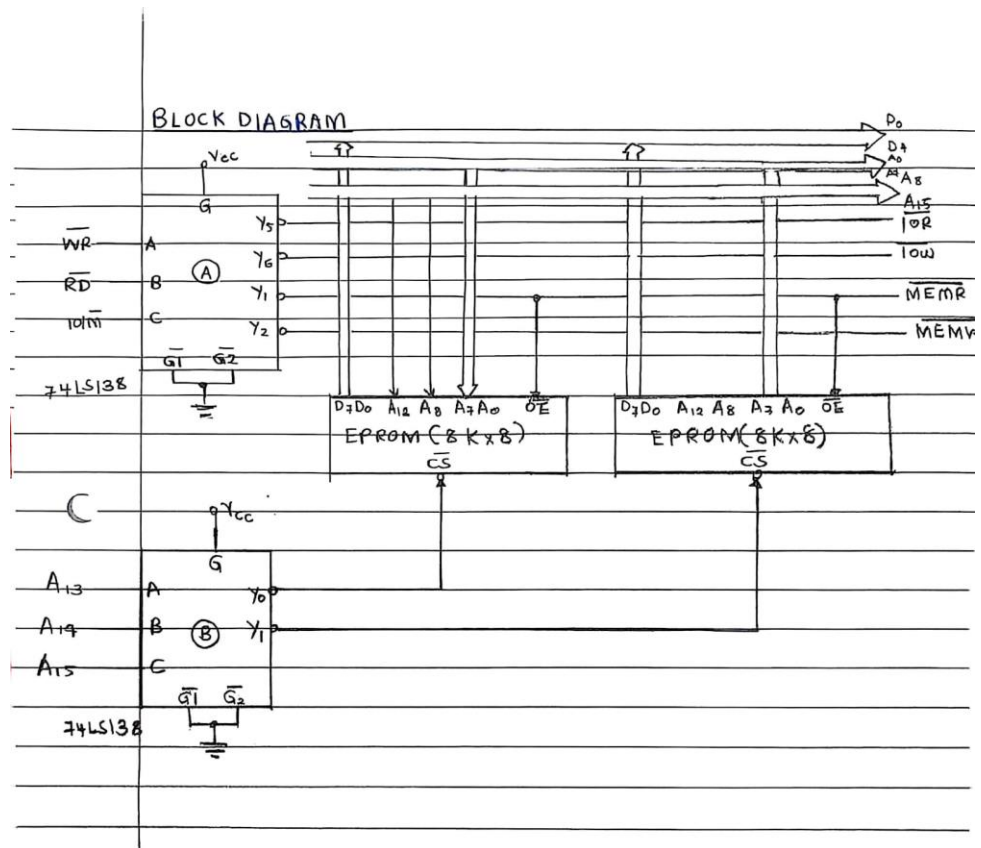
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On 2(e)

MEMORY MAP

Memory ICs	A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	Address
Starting address	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0000H
EPROM 1																	
End address	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1FFFH
EPROM 1																	
Starting address	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	2000H
EPROM 2																	
End address	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	3FFFH
EPROM 2																	

Final block diagram



Question 3

- a. Write an assembly language program to multiply 2 numbers using the Intel 8085 assembly codes.

Let the two numbers be 05H and 03H

Repeated addition:

MVI B, 05H ;Multiplicand

MVI C, 03H ;Multiplier

LOOP: ADD B ; A = A + B

DCR C

JNZ LOOP ; Repeat until C = 0

HLT

- b. List any 4 externally initiate control signals or interrupts of the Intel 8085 microprocessor.

TRAP (Non-maskable interrupt)

It is the highest priority hardware interrupt, typically reserved for critical system events.

RST 7.5 – maskable, vectored hardware interrupt with 2nd highest priority. It is edge-triggered.

RST 6.5 – maskable, level-triggered vectored interrupt

RST 5.5 - maskable, level-triggered vectored interrupt

RESET IN – resets program counter to zero and initializes the CPU.

READY – used to synchronize the microprocessor with slower peripheral devices.

INTR (Interrupt Request) – general-purpose, maskable hardware interrupt. It is level-triggered and non-vectored.

HOLD – indicates that a peripheral such as DMA controller is requesting use of the address and data bus.

- c. List any 4 functional grouping of the Intel 8085 instruction set.

Data transfer (copy) operations

Copies data from a location called a source to another location, called a destination, without modifying the contents of the source.

Arithmetic operations

Operations that perform mathematical calculations on numerical data stored in registers or memory.

Examples:

Addition, Subtraction, Increment/Decrement

Logical operations

Operations performed on binary data to manipulate bits or make decisions.

Examples:

AND, OR, Exclusive OR

Branching operations

Alter the sequence of program execution either conditionally or unconditionally.

Examples:

- Jump
- Call, Return & Restart

Machine control operations

These instructions control machine functions such as Halt, Interrupt or do nothing.

Question 4

- a. With the aid of an example EACH, illustrate any 2 categories of how Intel 8085 instructions are grouped in terms of the number of bytes.

One byte instruction

Consists of opcode only (1 byte).

Example: MOV A, B whose opcode is 78H which is 1 byte.

Two byte instruction

Consists with opcode + 8-bit data.

Example: MVI B, 02H

Opcode -> 06H

Data -> 02H

Three-byte instruction

Consists of opcode + 16 bit address

Example: JMP 6200H

Opcode -> C3H

16 bit address -> 6200H (2 bytes)

- b. Write a program which will store the contents of accumulator and flag register at 2000H and 2001H memory locations respectively.

Solution

LXI SP, 3000H ; Initialize stack pointer

PUSH PSW ; Push accumulator and flag register onto the stack

POP H ; L <- Flags, H <- Accumulator

MOV A, H ; Get accumulator

STA 2000H ; Store accumulator at 2000H

MOV A, L ; Get flag register

STA 2001H ; Store flags at 2001H

HLT

Explanation

PSW (Program Status Word) consists of:

- Accumulator (A)
- Flag Register (F)

So, when you execute: PUSH PSW, the 8085 saves both the accumulator and the flag register onto the stack.

Example

Suppose:

- SP = 3000H
- A = 5AH
- Flags = 86H

Before execution: SP = 3000H

After executing PUSH PSW:

- Address: 2FFFH; Data: 5AH (Accumulator)
- Address: 2FFE H; Data: 86H (Register)

SP = 2FFE H

The stack grows downwards, so the Stack pointer decreases by 2.

POP H -> It is an 8085 stack instruction that removes (pops) 2 bytes from the top of the stack and stores them in the HL register pair.

L gets the first byte (top of stack)

H gets the second byte.

Stack Pointer (SP) increases by 2.

STA (Store Accumulator) -> stores the contents of the Accumulator (A) into a specified memory location.

c. **Describe what is meant by peripheral mapped I/O scheme of the Intel 8085.**

⇒ Also known as **I/O mapped I/O** or **Isolated I/O**

Peripheral mapped I/O is an addressing scheme in Intel 8085 where peripheral devices are assigned a separate 8-bit I/O address space instead of memory addresses. The microprocessor communicates with these devices using the **IN** and **OUT** instructions. Since the I/O address is 8 bits wide, the 8085 can interface with 256 input/output ports. This scheme preserves the entire 64KB memory space for memory devices but limits the number of I/O ports and allows data transfer only through the Accumulator.

Summary

It treats them separately from memory.

I/O devices are assigned a "port number" within the 8-bit address range of 00H to FFH.

The user in this case would access these devices using the IN and OUT instructions only.

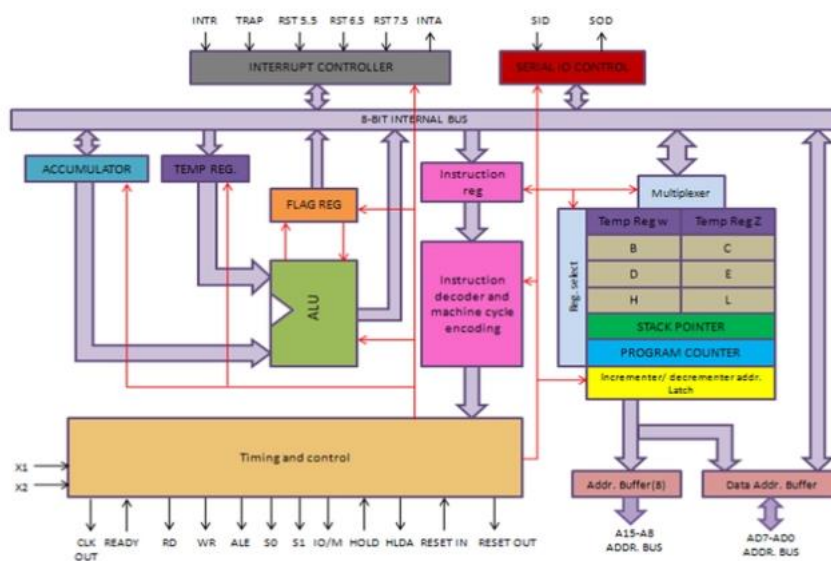
- 8-bit addressing
- Less decoder Hardware
- Can address $2^8 = 256$ locations
- Whole memory address space is available.
- Special instructions are used like IN, OUT
- Special control signals are used

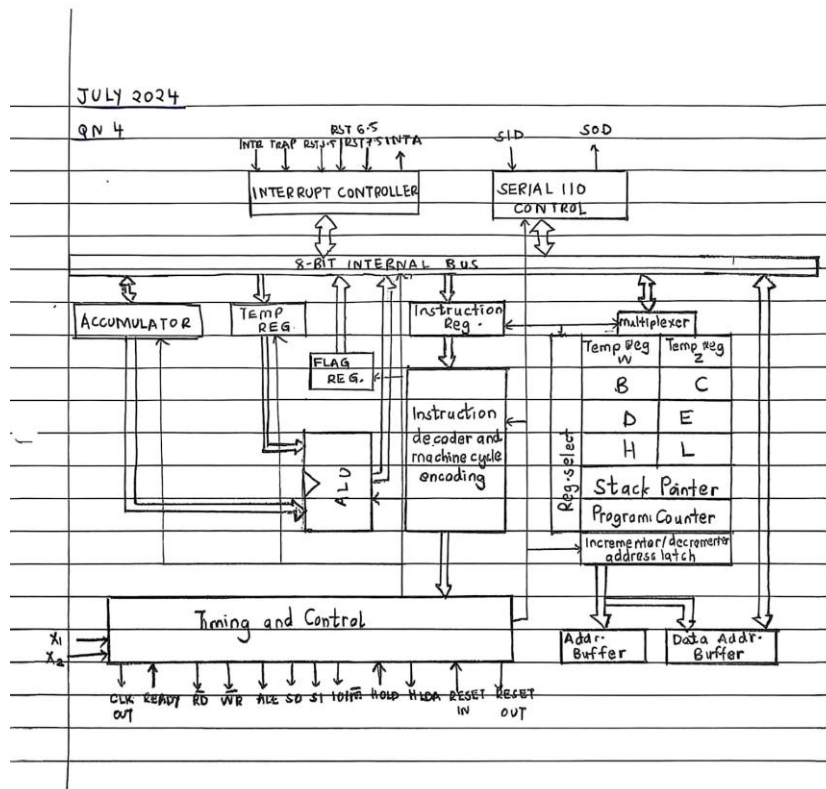
- Arithmetic & logic operations cannot be performed on data
- Data transfer between accumulator and I/O

July 2024

Question 4

- Using a well labelled diagram, illustrate the layout of the internal architecture of the 8085 microprocessor and explain the functions of the various sub-circuits.





Arithmetic & Logic Unit

Performs arithmetic & logical operations.

Consists of:

- Accumulator – stores one of the operands for ALU operations and usually receives the final result.
- Temporary register – used to hold the second operand during an ALU operation.
- Flag register – set of flip flops that include the status of the result.

Registers & Address Logic

General purpose registers: Pairs of 8-bit registers (B, C, D, E, H, L) used individually or as 16-bit pairs (BC, DE, HL) to hold data & memory addresses.

Stack pointer: 16-bit register that points to the top of the stack (R/W memory) in RAM.

Program Counter (PC): 16-bit register that holds the memory address of the next instruction to be executed.

Incrementer/Decrementer Address Latch: Adds or subtracts 1 from the contents of the Program Counter or Stack Pointer.

Instruction Unit

Handles the “fetch-decode-execute” cycle.

Instruction Register: Receives the opcode (instruction) from memory.

Instruction Decoder & Machine Cycle Encoding: Translates the opcode into specific electronic signals that tell the rest of the CPU what to do.

Control & Status Signals

Timing & Control – acts as the brain’s coordinator

Control signals – Includes RD (Read), WR (Write) and ALE (Address Latch Enable) to synchronize with memory and I/O devices.

Status signals: S0, S1 and IO/M indicate the type of operation being performed.

Specialized Controllers

Interrupt Controller: Manages external signals that “interrupt” the current program, such as TRAP (highest priority), RST 5.5, 6.5, 7.5 and INTR.

Serial I/O Control: Manages 1-bit serial communication through the SID (Serial Input Data) and SOD (Serial Output Data) pins.

Buses and buffers

8-bit internal bus: Highway that connects all internal components.

Address/Data buffers: The 8085 uses a multiplexed bus. The lower 8 bits (AD0 – AD7) carry both addresses and data at different times while the upper 8 bits (A8 – A15) carry only addresses.

- b. Consider that you have two numerical data values stored in an external memory interfaced to a microprocessor (8085). One value is stored in a memory location whose address is 3000H, and the second value is stored in a memory location whose address is 3001H. Write an Assembly Language Program (ALP) that will load these data values into any of the microprocessor registers, add them in the ALU and output the results from the accumulator to an external memory location whose address is 3002H. Ensure your ALP constitutes of the memory addresses, opcode, Labels (if any), Mnemonics, and Remarks, refer to the attached instruction set for the opcodes.

Address	Opcode	Label	Mnemonic	Remarks
2000H	3A		LDA 3000H	Load contents of 3000H into accumulator
2001H	00			Low-order address byte
2002H	30			High-order address byte
2003H	47		MOV B, A	Copy first number from A to B
2004H	3A		LDA 3001H	Load contents of 30001H into accumulator
2005H	01			Low-order address byte
2006H	30			High-order address byte

2007H	80		ADD B	Add contents of B to accumulator
2008H	32		STA 3002H	Store result from accumulator to 3002H
2009H	02			Low-order address byte
200AH	30			High-order address byte
200BH	76		HLT	Stop program execution

Compact ALP

LDA 3000H

MOV B, A

LDA 3001H

ADD B

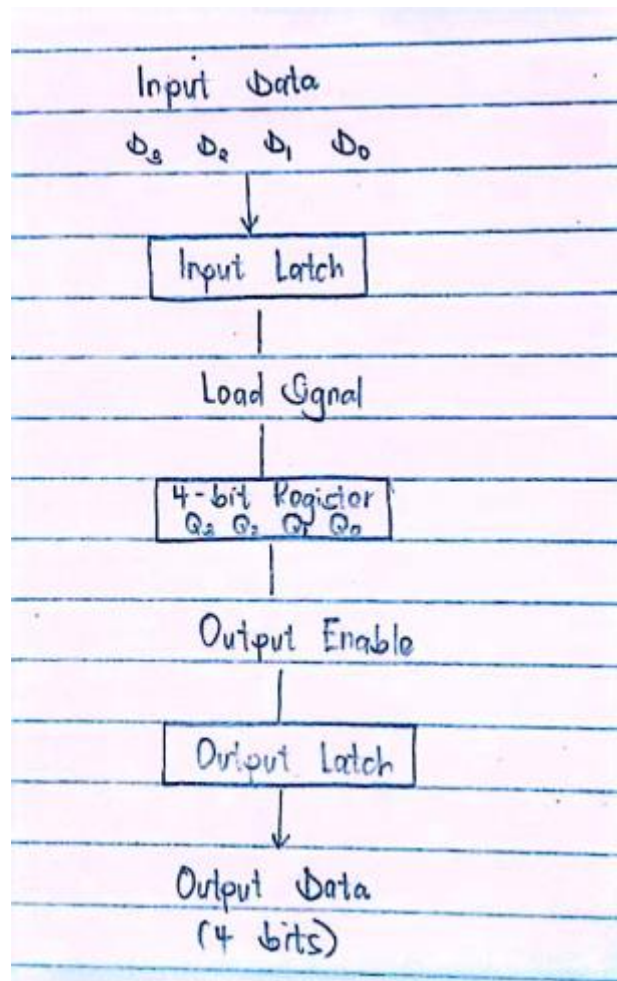
STA 3002H

HLT

January_2022

Question 2

- With the aid of a diagram, illustrate how an input and output latch are used to control data flow in a four-bit register memory unit.



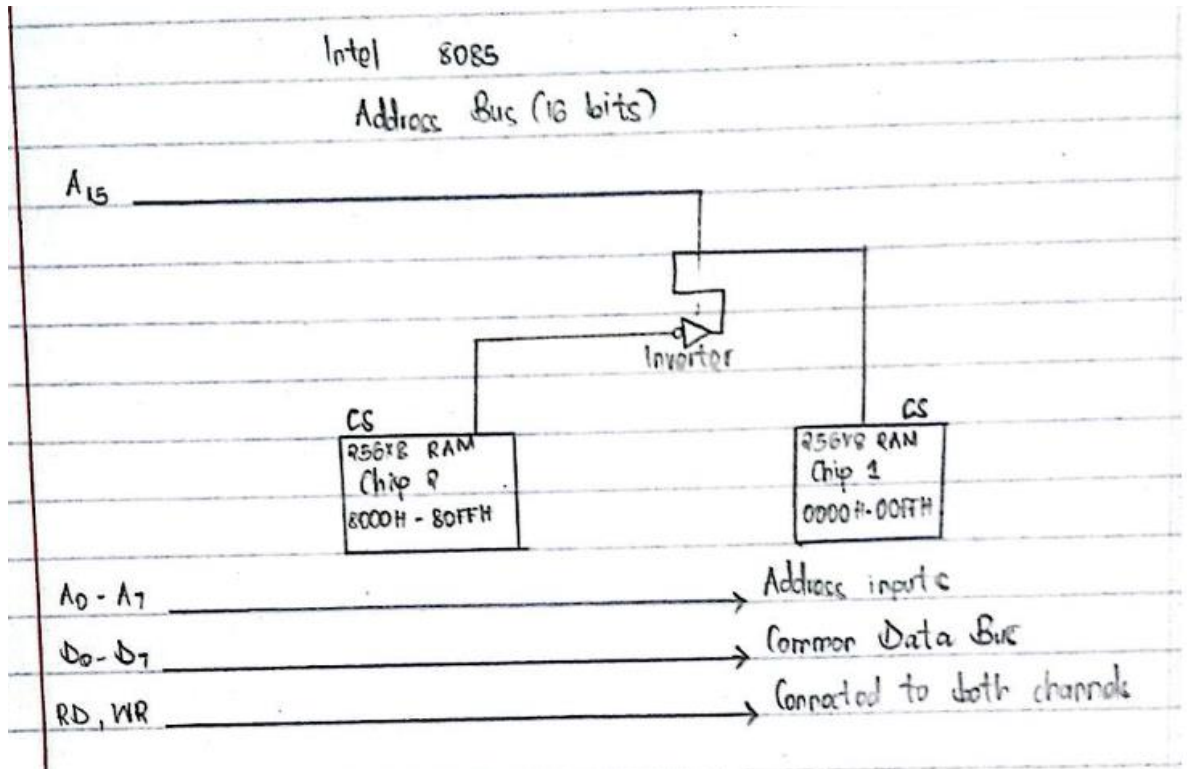
- The input latch temporarily stores incoming 4-bit data before it is transferred into the register.
- The load signal determines when the register accepts the data.
- The 4-bit register stores the data until it is changed.
- The output latch isolates the register from the output bus and releases data only when the output enable signal is active
- The latch prevents unwanted changes and ensures controlled data transfer.

b) Assume there is a requirement to implement a 512 x 8 memory block using two 256 x 8 memory blocks. Also assume the two memory units available must be implemented using address 0000H and 8000H ranges only. Using appropriate diagram show how this memory should be implemented if there was no option of using decoders for chip selection.

- Required memory: 512 x 8
- Available chips: 256 x 8

Memory Chips Required = $512/256 = 2$ chips

Memory Chip	Starting Address	Ending Address	Chip Select
Chip 1	0000H	00FFH	A15=0
Chip 2	8000H	80FFH	A15=1



- Address lines A₀-A₇ are connected to both memory chips because each chip stores 256 locations
- The data bus (D₀-D₇) is connected in parallel to both chips. The $\overline{RD}$ and $\overline{WR}$ control signals are common to both memory chips.
- Since no decoder is allowed, the A₁₅ address line is used directly for chip selection:
 - A₁₅ = 0** selects the chip beginning at 0000H (through an inverter because most memory chips have an active-low $\overline{CS}$ input).
 - A₁₅ = 1** selects the chip beginning at 8000H.
- The remaining higher address lines (A₈-A₁₄) are ignored. This is an example of **partial address decoding**, which results in address mirroring.

c) State the FOUR steps involved during an opcode fetch operation in the Intel 8085 assuming data 4FH is fetched from location 2008H.

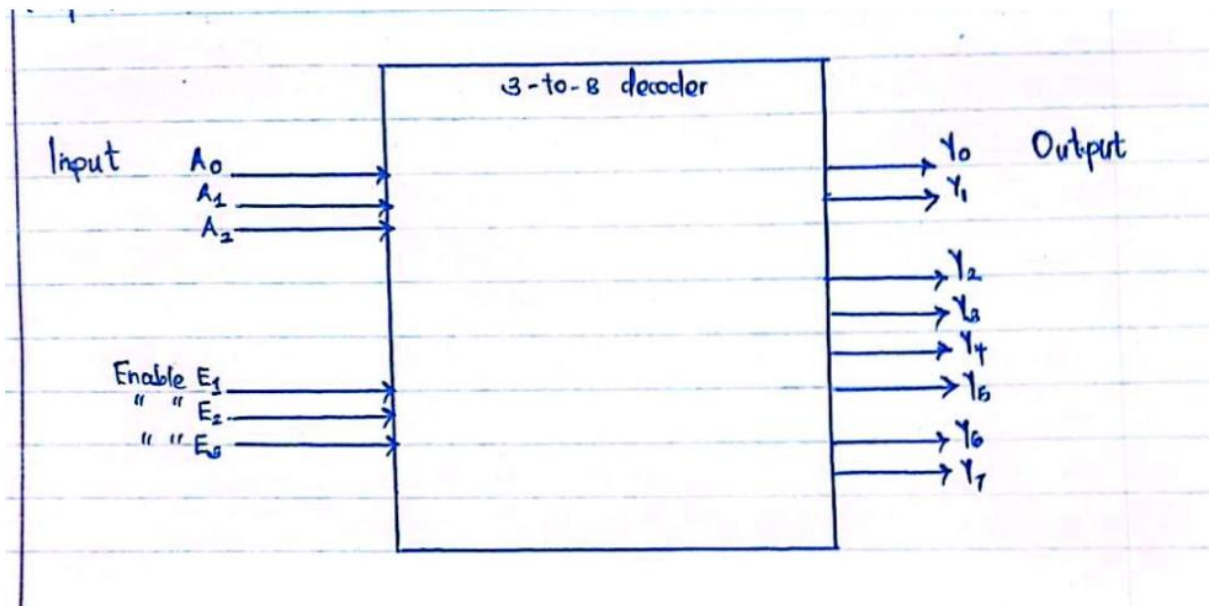
- i) The Program Counter (PC) places address 2008H on the address bus.
- ii) The processor activates $\overline{RD}$ to read memory
- iii) Memory location 2008H places 4FH onto the data bus.
- iv) The processor loads 4FH into the Instruction Register, increments the Program Counter to 2009H, and decodes the instruction.

d) Describe any three differences between a peripheral mapped I/O and a memory mapped I/O.

Peripheral Mapped I/O	Memory Mapped I/O
i) Accessed using IN and OUT instructions only	i) Accessed using memory instructions such as MOV, LDA, STA
ii) Uses separate I/O address space	ii) Uses normal memory address space
iii) Supports only 256 ports	iii) Supports up to 64 KB address space

Question Three

a) Draw a 3-to-8 decoder and explain why it is called a tri-state device.



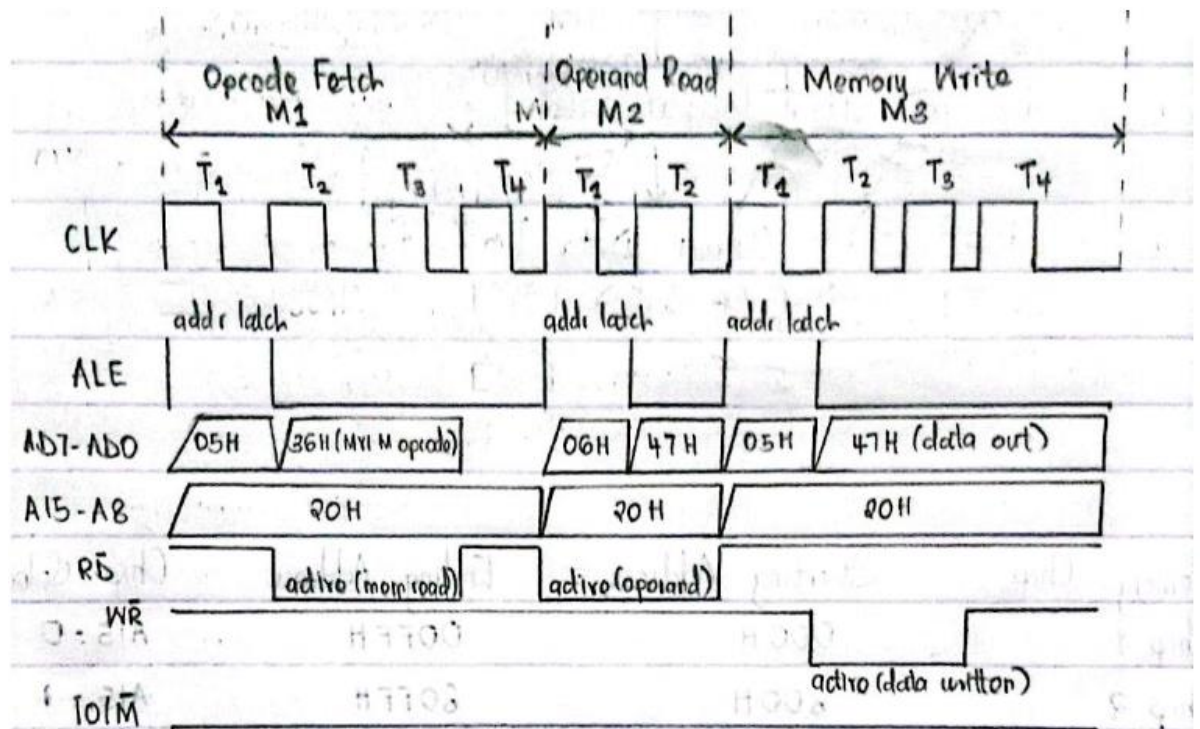
- A tri-state device is a digital circuit component that produces three distinct output conditions in our case the 3-to-8 decoder produces:

- i) **Logic 0 (Low)** – The output connected to the ground reference
- ii) **Logic 1 (High)** – The output connected to the positive supply voltage
- iii) **High-Impedance** – It disconnects the output from the bus, allowing multiple devices to share the same data bus without causing conflicts.

b) What is the function of the following control signals.

- i) **ALE** – Separates the lower-order address from multiplexed address/data bus (AD0-AD7)
- ii) **$\overline{RD}$** - Indicates that the processor is reading data; causes memory or an I/O device to place data on the data bus.
- iii) **$\overline{WR}$** - Indicates that the processor is writing data; Enables memory or an I/O device to store data from the data bus.
- iv) **$\overline{IO/M}$** - Distinguishes between memory and I/O operations; **$\overline{IO/M}(0)$** - Memory Operation and **$\overline{IO/M}(1)$** - I/O operation.

c) With the aid of a well labelled TIMING diagrams illustrate how data 47H can be written to location 2005H.



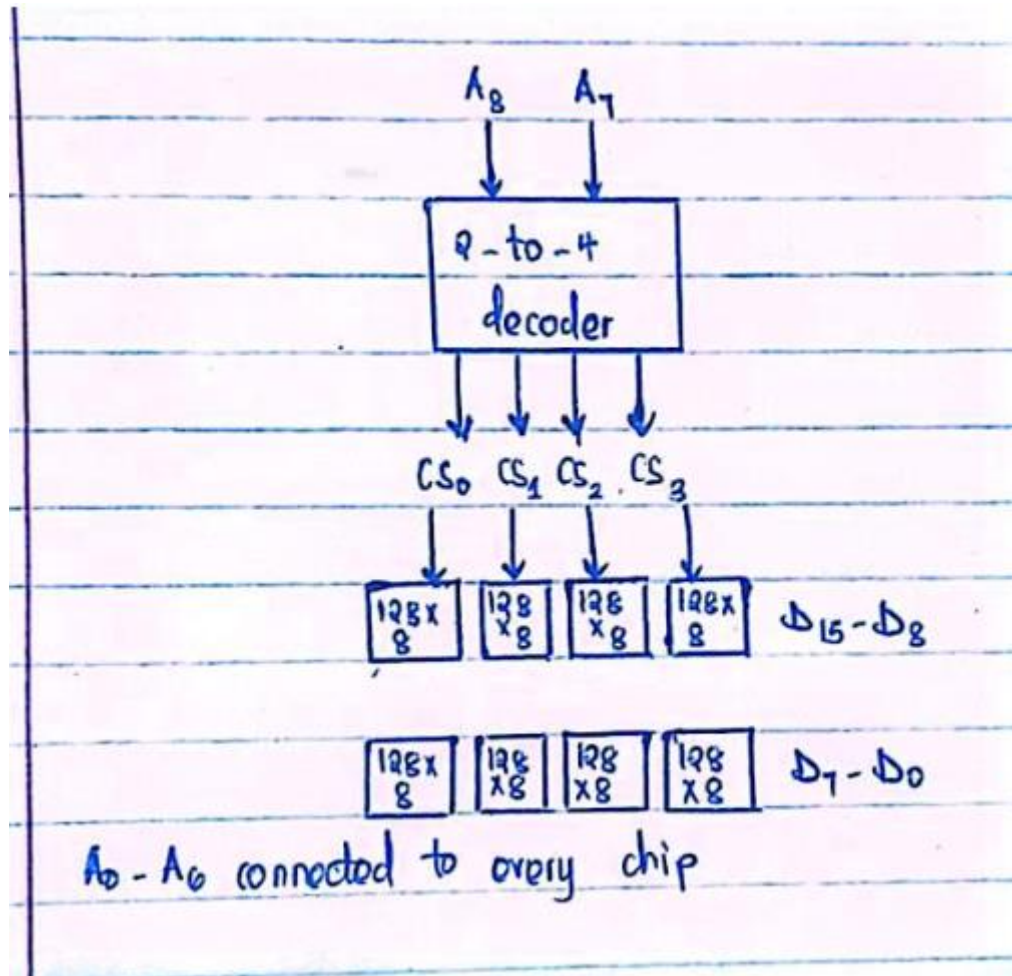
- **Instruction:** `MVI M, 47H` | **Address:** 2005H | **Data:** 47H | 10 T-states, 3 machine cycles
- M1 (T1–T4):** Fetch opcode 36H from PC (addr 2005H). ALE latches 05H. RD active T2–T3.
- M2 (T1–T2):** Read immediate data 47H from next address (2006H). RD active T2.
- M3 (T1–T4):** Write 47H to HL pair address (2005H). ALE latches 05H. WR active T2–T3. IO/M stays LOW — confirms all three cycles are memory (not I/O) operations. AD7–AD0 carries addr low byte in T1 (latched on ALE rising edge), then data from T2.

April 2024.

Question Two.

- You are tasked with designing a memory block measuring 512x16, but due to budget constraints, your company can only produce memory chips sized 128x8. Explain how you would achieve the required memory block using the available components.
- Required Memory: 512x16
 - Available chip: 128x8

- Memory depth: $512/128 = 4$ banks
- Word size: $16/8 = 2$ chips per bank
- Therefore: $4 \times 2 = 8$ memory chips



- Two chips operate in parallel to produce a 16-bit word.
- Four banks provide 512 memory locations.
- A_0-A_6 address each chip.
- A_7-A_8 are decoded to select one bank.

b) Describe the three functional bus of the Intel 8085 microprocessor

i) Address Bus.

- It carries the address of memory and I/O devices and can address 64 KB of memory.
- It is 16 bits wide (A_0-A_{15}) and unidirectional.

ii) **Data Bus.**

- It transfers data between the processor and the memory or I/O devices.
- It is 8 bits wide(D0-D7) and it is bidirectional.

iii) **Control Bus.**

- It carries control and timing signals.
- It controls memory read/write and I/O operations for example; RD, WR, ALE.

c) **Describe the function of the following Intel 8085 signals.**

- i) **INTA** – generated by the processor after accepting an INTR interrupt and informs the interrupting device that its request has been accepted; it enables the external device to place the interrupt instruction on the data bus.
- ii) **TRAP** – it is the highest-priority, non-maskable interrupt and it is used for emergency conditions such as power failure.
- iii) **HOLD** – used by DMA devices to request control of the system buses from the processor and after completing the current machine cycle, the processor releases the buses.
- iv) **READY** – indicates whether memory or an I/O device is ready; READY (1) = processor continues execution and READY (0) = processor enters wait states until READY becomes high

d) **To simplify microprocessor design, certain control signals are derived by combining primary signals logically. Explain the generation process for the following signals:**

i) **MEMR**

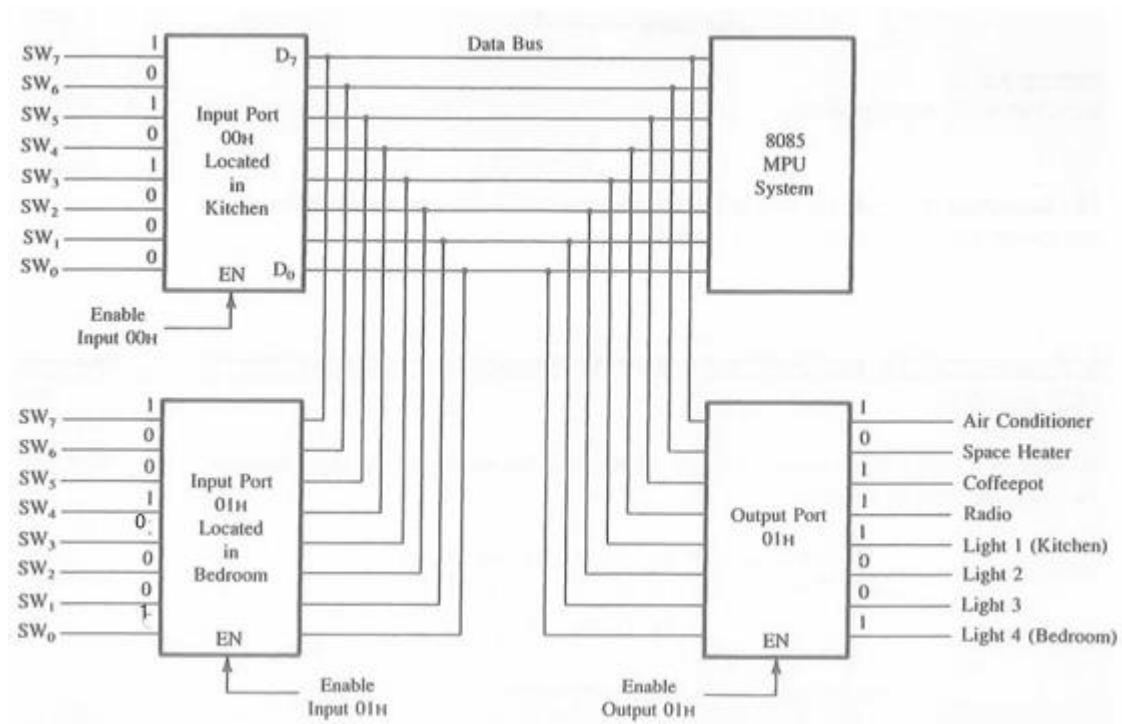
- Memory Read is generated when $IO/\overline{M} = 0$ and $R\overline{D} = 0$ hence $MEMR = R\overline{D} + IO/\overline{M}$

ii) **IOR**

- I/O Read occurs when $IO/\overline{M} = 1$ and $\overline{RD} = 0$ hence $\overline{IOR} = \overline{RD} + (IO/\overline{M})'$

Question Three.

- a) Utilizing an Intel 8085 microprocessor to manage a home appliance system, represented in Fig. 2 below, involves two input ports (Port 00H and Port 01H) linked to switches, and one output port associated with actuators. The bit pattern at the input port reflects the switch condition, with '1' indicating ON and '0' indicating OFF. Provide a simple assembly program to ensure each appliance operates in accordance with the input port condition.



```

; =====
; HOME APPLIANCE CONTROL SYSTEM FOR 8085
; =====
; Input Ports: 00H (Kitchen), 01H (Bedroom)
; Output Port: 01H (Actuators)
; =====

```

```

ORG 8000H    ; Starting address

```

```

START: IN 00H      ; Read Kitchen switches from Port 00H
      MOV B, A      ; Save Kitchen switch pattern in B register

      IN 01H        ; Read Bedroom switches from Port 01H
      MOV C, A      ; Save Bedroom switch pattern in C register

; Combine both port readings for output
      MOV A, B      ; Get Kitchen switches
      ANI 0FH       ; Mask lower 4 bits (Kitchen appliances)
                        ; Bits 0-3 = Light 1, Light 2, Light 3, Light 4

      MOV D, A      ; Store Kitchen controlled bits

      MOV A, C      ; Get Bedroom switches
      ANI 0FH       ; Mask lower 4 bits
      RLC           ; Rotate left 4 times to move to upper nibble
      RLC           ; Now bits 0-3 become bits 4-7
      RLC
      RLC
      ORA D         ; Combine with Kitchen bits

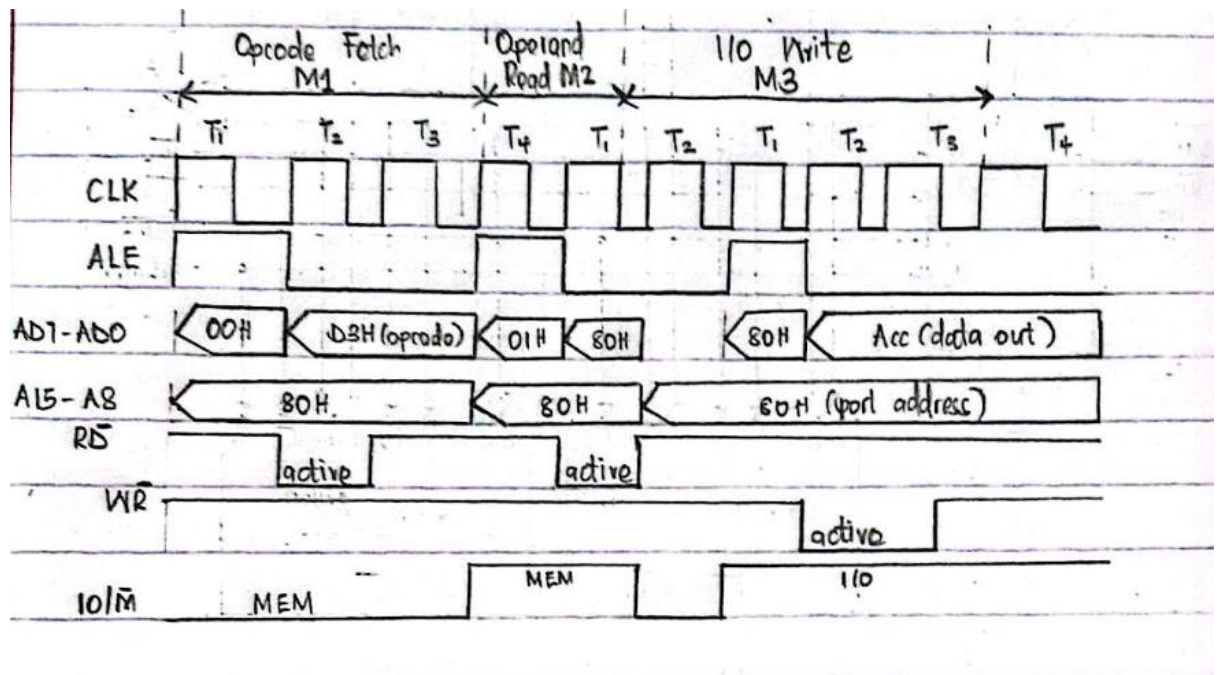
      OUT 01H       ; Send to Output Port (Actuators)

      JMP START     ; Continuous loop to update outputs

      HLT           ; Halt (optional - not reached)

```

- b) The following instructions are executed by the Intel 8085 microprocessor. Draw the complete timing diagram that will ensure the opcodes are correctly executed: 8000: OUT 80H

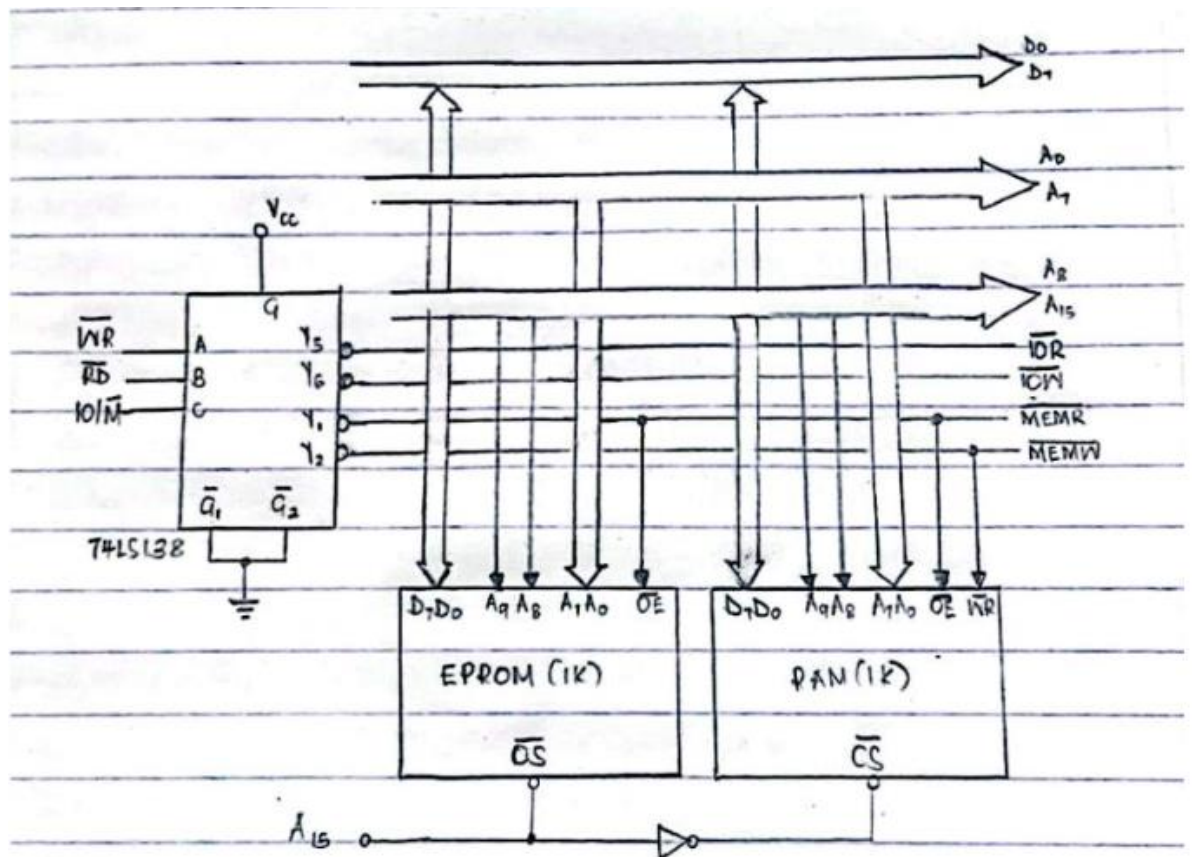


- Instruction: `OUT 80H` | Opcode: D3H | Operand: 80H | 10 T-states, 3 machine cycles
- M₁ (T₁-T₄): Fetch opcode D3H from address 8000H. ALE high at T₁, RD active T₂-T₃.
- M₂ (T₁-T₂): Read port address 80H from address 8001H. ALE high at T₁, RD active T₂.
- M₃ (T₁-T₄): Write accumulator to I/O port 80H. ALE high T₁, IO/M high, WR active T₂-T₃. AD7-AD0 carries low address byte in T₁ (latched by ALE), then data in T₂ onwards.

c) With the aid of an appropriate diagram describe the following I/O encoding techniques.

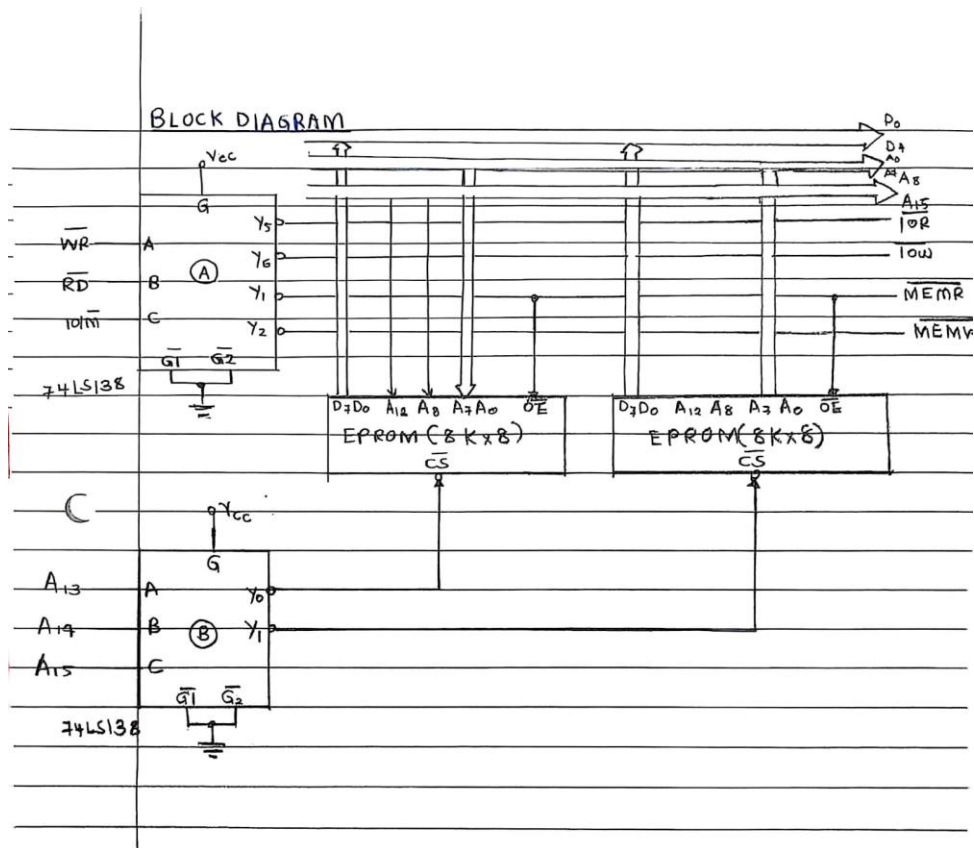
i) Partial encoding

- This is where decoding logic can be eliminated by using individual high-order address lines to select memory chips.



ii) Absolute encoding

- All the higher address lines are decoded to select the memory chip, and the memory chip is selected only for the specified logic level on these high-order address, no other logic levels can select the chip.



d) Explain why an interrupt may be considered maskable and vectored in the context on Intel 8085 interrupts.

- A maskable interrupt is one that can be enabled or disabled by a software. The processor can ignore these interrupts when executing critical sections of a program **for example; INTR** while the vectored interrupt has a predefined memory address where its interrupt service routine begins. When the interrupt is accepted, the 8085 automatically jumps to this fixed address without requiring the external device to provide it for example;

Interrupt	Vector Address
TRAP	0024H

July 2024.

Question Three.

- a) Briefly describe the historical trends in evolution of intel microprocessors. Use a table for the different key performance indicators.

Processor	Year	Word Size	Address Bus	Clock Speed	Key Features
Intel 4004	1971	4-bit	12-bit	740 kHz	First commercial processor
Intel 8008	1972	8-bit	14-bit	0.8 MHz	Improved instruction set
Intel 8080	1974	8-bit	16-bit	2 MHz	Faster processing and larger memory
Intel 8085	1976	8-bit	16-bit	3-6 MHz	Single, +5V supply, built-in interrupts
Intel 8086	1978	16-bit	20-bit	5-10 MHz	1 MB memory addressing
Intel 80286	1982	16-bit	24-bit	6-25 MHz	Protected mode
Intel 80386	1985	32-bit	32-bit	16-40 MHz	Virtual memory and multitasking

- Word size increased from 4-bit to 32-bit
- Memory addressing capacity expanded
- New processors introduced multitasking, protected mode, virtual memory, and improved performance.
- Clock speeds steadily increased.

b) Considering the Intel 8085 instruction set, write an Assembly Language Program (ALP) that loads the hexadecimal numbers 9BH and A7H in the registers D and E, respectively, and adds the numbers. If the sum is greater

than FFH, display 01H at PORT0; otherwise display the sum. Ensure your ALP constitutes of the memory addresses, machine code, Labels, Mnemonics, and Remarks refer to the attached instruction set for the machine codes

- $9BH + A7H = 142H$ this means a carry is generated

Address	Machine Code	Label	Mnemonic	Remarks
2000H 2001H	16 9B	START	MVI D, 9BH	Load the 8-bit immediate hexadecimal data 9BH into register D
2002H 2003H	1E A7		MVI E, A7H	Load the 8-bit immediate hexadecimal data A7H into register E
2004H	7A		MOV A, D	Transfer the contents of register D(9BH) into the accumulator
2005H	83		ADD E	Add register E to the Accumulator. Result= 42H. Carry flag resets to 1
2006H 2007H 2008H	D2 0DH 20H		JNC DISP_SUM	Jump to label DISP_SUM (200DH) if Carry =0 (Sum <= FFH)
2009H 200AH	3E 01		MVI A, 01H	Carry is active (Sum > FFH).

				Explicitly override Accumulator with error value 01H.
200BH 200CH	D3 00		OUT PORT 0	Write the error value 01H from the Accumulator to peripheral PORT0.
200DH	76		HLT	Halt program execution path for overflow scenario.
200EH 200FH	D3 00	DISP_SUM:	OUT PORT 0	Alternate branch destination. Send normal, non-overflow sum to PORT0.
2010H	76		HLT	Halt standard program execution path .